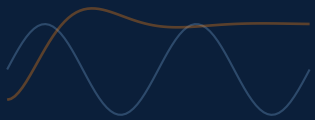
WEEK 1 · DAY 3 / 5

Mechatronics and Robotics Engineering (50 h)



Control in Robotic & Mechatronic Systems

Block diagrams & state space ▪ Bode plots ▪ PID control ▪ Stability & performance

Mechatronics and Robotics Engineering Programme

Week 1 · Day 3 ▪ 2 weeks · 10 days · 50 hours

Agenda and learning outcomes

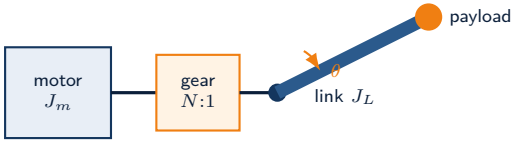
Code	Session	Time
2.3	Block diagrams, feedback and state-space basics	1.0 h
2.4	Frequency response and Bode plots	1.0 h
2.5	PID control: P/I/D effects, tuning, windup, feedforward	1.5 h
2.6	Stability and performance: specs, steady-state error, margins, bandwidth	1.5 h
	Net teaching time (+ short breaks)	**5.0 h**

Running example all day: **one robot joint** (motor + gearbox + link). Refresher boxes: matrices, sine response, decibels. Small grey "Going further" items are optional.

By the end of today you can ...

- **draw** a feedback loop as a block diagram and **reduce** it to one transfer function
- **write** a simple system in state-space form
- **read** a Bode plot (dB, degrees) and **spot** a resonance
- **explain** what P, I and D do to a robot joint and **pick** starting gains
- **judge** a loop from overshoot, steady-state error, phase/gain margin and bandwidth

What we learned so far (1/2): the robot joint hardware



amplifier sends current $i \Rightarrow$ torque $k_t i$

$$J_{\text{eq}} = J_m + \frac{J_L}{N^2}$$

From W1D1–W1D2:

- Electric motors make torque $T = k_t\, i$; back-EMF $e = k_e\, \omega$.
- Gearboxes trade speed for torque; the load inertia is seen **divided by** N^2 at the motor.
- Backlash and elasticity (harmonic drive!) make the joint less "perfect".
- Sizing: peak and RMS torque, speed, thermal limits.

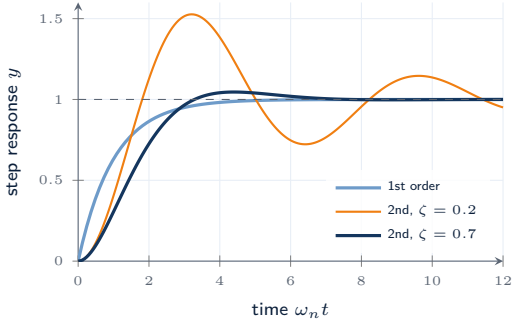
Today we ask: **how do we make this joint go exactly where we want?**

What we learned so far (2/2): models and transfer functions

- Physics ⇒ ODE (mass–spring–damper, motor).
- Laplace: ODE ⇒ algebra; $G(s) = Y(s)/U(s)$.
- **Poles** = roots of the denominator: they decide stability and speed.
- First order: K, τ (63 % at $t = \tau$).
- Second order: ω_n, ζ (bounciness).

$$G_{\text{joint}}(s) = \frac{\Theta(s)}{T(s)} = \frac{1}{Js^2 + Bs}$$

Torque in, angle out. Today: $J = 0.5\,\text{kg m}^2$, $B = 0.1\,\text{Nm s/rad}$ (joint side).



Same speed scale, very different shapes: ζ sets the overshoot.

Session 1

2.3 Block Diagrams, Feedback and State-Space Basics

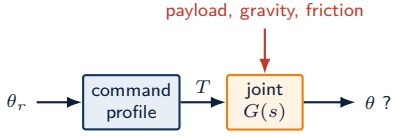
Why does a robot need feedback at all?



Oleg Yunakov, CC BY-SA 4.0, via Wikimedia Commons

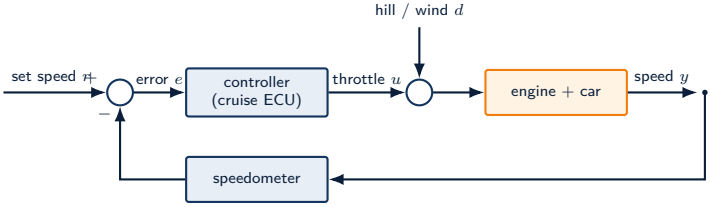
An industrial arm repeats a pose to about ± 0.05 mm while payload, temperature and friction change.

Open loop: send a fixed torque profile and hope.



- The joint is a double integrator: a tiny torque error grows into a **large drift** in angle.
- The model is never exact (payload unknown, friction varies, gear wears).
- **Closed loop**: measure θ with an **encoder**, compare with θ_r, correct the torque.

Feedback: the cruise-control story



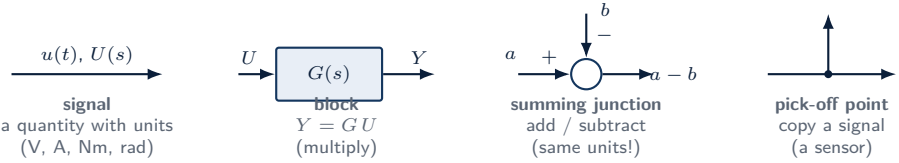
- **Measure** the speed, **compare** with the set speed, **correct** the throttle.
- Robot joint: set speed → target angle, speedometer → encoder, throttle → motor current.
- Feedback fights disturbances **it does not know about**.

Worked example: P control only

Controller $u = K_p\, e$. Car: 2 km/h per % throttle ($G = 2$); $K_p = 4.5\,\%/(\text{km/h})$ ⇒ loop gain $L = K_p G = 9$.

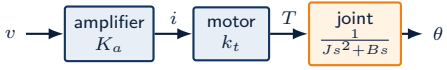
1. $y = \frac{L}{1+L}\, r = \frac{9}{10} \cdot 100 = 90\,\text{km/h}$
2. A **10 km/h offset** remains: P alone needs an error to push.

Block diagrams: the four building blocks



A **block diagram** is a map of how signals flow and change. Each block is a transfer function from Day 2; the arrows are signals.

- Arrows have one direction: cause $\rightarrow$ effect.
- Feedback = an arrow going **back** to an earlier summing junction.



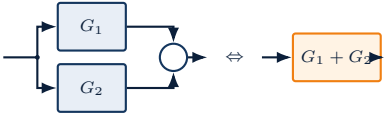
Series blocks multiply: $\Theta/V = K_a\,k_t/(Js^2 + Bs)$.

Block-diagram algebra

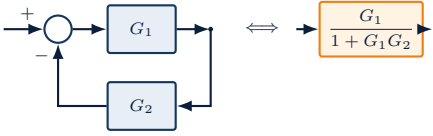
Series



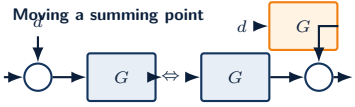
Parallel



Negative feedback



Moving a summing point

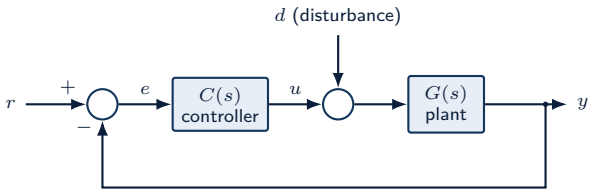


Why feedback is powerful

$G_1 = 10$, $G_2 = 1$: $10/11 = 0.91$. Double G_1 to 20: $20/21 = 0.95$.
A 100 % change in G_1 changes the closed loop by only **4 %**.

Feedback formula: "forward over one plus loop".
Going further: Mason's gain rule for tangled diagrams; positive feedback gives $G_1/(1 - G_1G_2)$.

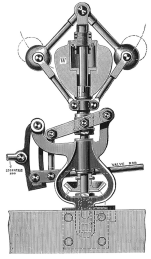
The standard feedback loop and its two key functions



Loop gain $L(s) = C(s)G(s)$: go once around the loop. **Sensitivity** S: what is left of a disturbance; **complementary sensitivity** T: how well y follows r.

$$S = \frac{1}{1+L}, \qquad T = \frac{L}{1+L}$$

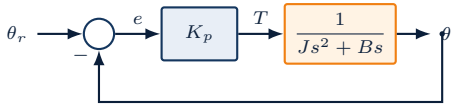
- Cruise control: $L = 9 \Rightarrow S = 0.1,\ T = 0.9;\ S + T = 1$ always.
- Big $|L| \Rightarrow$ small $S \Rightarrow$ good tracking – **but** sensor noise also reaches y.



Heat Engines (1913), scan by Andy Dingley, Public domain, via Wikimedia Commons

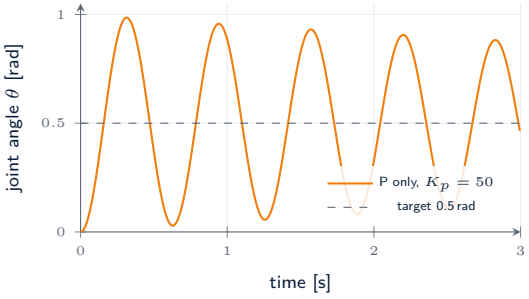
Watt's centrifugal governor: a mechanical P controller for steam-engine speed. Maxwell's paper "On Governors" (1868) started control theory.

Worked example: closing a P loop around the robot joint



$$\frac{\Theta}{\Theta_r} = \frac{K_p}{Js^2 + Bs + K_p}$$

1. Compare with $\omega_n^2/(s^2 + 2\zeta\omega_n s + \omega_n^2)$.
2. $K_p = 50\,\text{Nm/rad}$:
   $\omega_n = \sqrt{K_p/J} = \sqrt{100} = 10\,\text{rad/s}$.
3. $\zeta = B/(2\sqrt{K_p J}) = 0.1/10 = 0.01$.
4. Almost no damping $\Rightarrow$ the arm **rings** for many seconds.



Feedback alone does not guarantee **good** behaviour: the loop must be designed (sessions 2.5, 2.6).

Refresher: matrices for state space

Refresher: matrix times vector

Each row of A times the vector $\boldsymbol{x}$ gives one entry:

$$\begin{bmatrix} 0 & 1 \\ -4 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \cdot 1 + 1 \cdot 2 \\ -4 \cdot 1 - 2 \cdot 2 \end{bmatrix} = \begin{bmatrix} 2 \\ -8 \end{bmatrix}$$

Why? Two coupled ODEs

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = -4x_1 - 2x_2$$

fit in one line: $\dot{\boldsymbol{x}} = A\boldsymbol{x}$ with the matrix above.

Eigenvalues = the natural "speeds" of A:

$$\det(\lambda I - A) = 0$$

1. $\lambda I - A = \begin{bmatrix} \lambda & -1 \\ 4 & \lambda + 2 \end{bmatrix}$
2. $\det = \lambda(\lambda + 2) + 4 = \lambda^2 + 2\lambda + 4$
3. Roots: $\lambda = -1 \pm j1.73$

Eigenvalues of A = poles of the system. This is a mass–spring–damper with $m = 1$, $k = 4$, $c = 2$ ($\omega_n = 2$, $\zeta = 0.5$).

State-space representation

State $\boldsymbol{x}$: the few numbers that describe "where the system is now" (positions, speeds, currents).

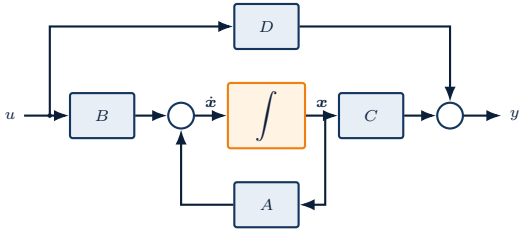
$$\dot{\boldsymbol{x}} = A\boldsymbol{x} + Bu, \qquad y = C\boldsymbol{x} + Du$$

Robot joint, $x_1 = \theta$, $x_2 = \omega = \dot{\theta}$, $u = T$:

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{B}{J} \end{bmatrix}, \; B = \begin{bmatrix} 0 \\ \frac{1}{J} \end{bmatrix}, \; C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

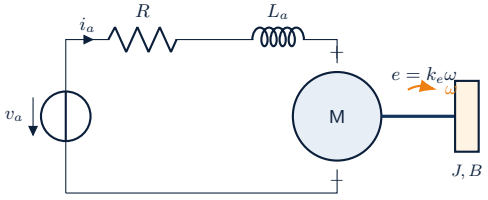
- Handles many inputs/outputs (a 6-axis arm!); $D = 0$ here.
- Basis for observers and the Kalman filter (W2D5).

Going further: $G(s) = C(sI - A)^{-1}B + D$; states are not unique.



One integrator per state variable: integrators "remember" the state. Eigenvalues of the joint's A: 0 and $-B/J = -0.2\,\mathrm{s}^{-1}$ (the same as the poles of $1/(Js^2 + Bs)$).

The joint motor in both representations



$$L_a \frac{di_a}{dt} = v_a - Ri_a - k_e\,\omega \quad \text{(KVL)}$$

$$J \frac{d\omega}{dt} = k_t\, i_a - B\omega \quad \text{(Newton)}$$

SI units: k_t [Nm/A] $= k_e$ [V s/rad].

State space: two energy stores $\Rightarrow$ two states $\boldsymbol{x} = [i_a,\ \omega]^\top$, $u = v_a$, $y = \omega$:

$$A = \begin{bmatrix} -\frac{R}{L_a} & -\frac{k_e}{L_a} \\ \frac{k_t}{J} & -\frac{B}{J} \end{bmatrix},\ B = \begin{bmatrix} \frac{1}{L_a} \\ 0 \end{bmatrix},\ C = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

Transfer function (Laplace of both equations, eliminate I_a):

$$\frac{\Omega(s)}{V_a(s)} = \frac{k_t}{(L_a s + R)(Js + B) + k_t k_e}$$

- The $k_t k_e$ term is **back-EMF**: the motor has a **built-in feedback loop** (faster $\Rightarrow$ less current).
- Load torque T_L (gravity, payload) is a second input: a disturbance.

Worked example: a small joint servo motor

Motor: $R = 1.2\,\Omega$, $L_a = 2.4\,\text{mH}$, $k_t = k_e = 0.1\,\text{Nm/A}$, $J = 2 \times 10^{-4}\,\text{kg m}^2$ (motor + reflected link), $B = 10^{-5}\,\text{Nm s/rad}$.

1. Denominator divided by $L_a J$:

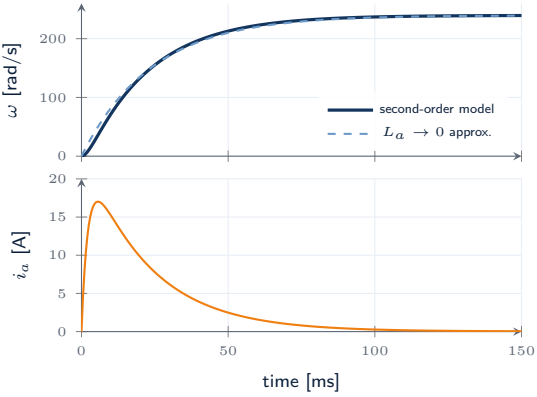
$$s^2 + 500\,s + 20\,860$$

2. Eigenvalues/poles: $-46\,\text{s}^{-1}$ (mechanical, slow), $-454\,\text{s}^{-1}$ (electrical, fast).

3. DC gain: $G(0) = \dfrac{k_t}{RB + k_t k_e} \approx 10\,\text{rad/s}$ per V.

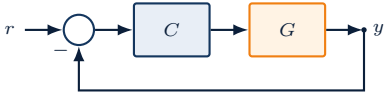
4. 24 V step $\Rightarrow \approx 240\,\text{rad/s}$ (2290 rpm), settles in $\approx 4/46 = 87\,\text{ms}$.

Starting current $\approx 24\,\text{V}/1.2\,\Omega = 20\,\text{A}$: a reason for current control (W1D5).



24 V voltage step, no load; simulated from the state-space model.

Session 2.3 – Check your understanding



Check your understanding

1. Reduce the loop on the right to one transfer function Y/R for $C = 4$ and $G = \frac{1}{s+1}$. What is the final value for a unit step?
2. In the P-controlled joint, what happens to ω_n and ζ if the payload doubles J?
3. How many state variables does the DC motor have, and why?

Session 2.3 – Summary and key terms

Key take-aways

- Feedback = measure, compare, correct; it fights unknown disturbances.
- Block diagrams: series multiply, parallel add, feedback gives $G_1/(1 + G_1G_2)$.
- $S = 1/(1 + L)$ and $T = L/(1 + L)$ describe every feedback loop.
- State space $\dot{\boldsymbol{x}} = A\boldsymbol{x} + Bu$: eigenvalues of A = poles.
- P control of an inertia gives an **undamped** spring: we need more (D).

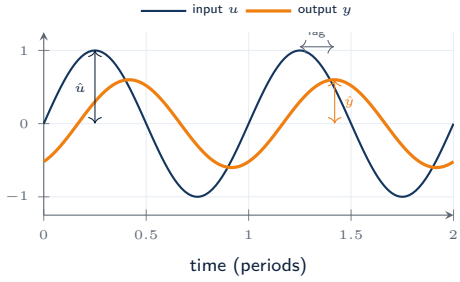
Key terms

Open / closed loop: without / with measurement.
Summing junction: adds or subtracts signals.
Loop gain $L = CG$: gain once around.
Sensitivity S: fraction of disturbance left.
State: variables that store energy/memory.

Session 2

2.4 Frequency Response and Bode Plots

Refresher: what a sinusoidal frequency response means



Wave a robot arm slowly: the tip follows. Wave it fast: the tip moves less and **lags behind**.

Refresher: frequency response

A linear stable system driven by a sine answers with a sine of the **same frequency**, only scaled and shifted.

$$y = |G(j\omega)|\,\hat{u}\,\sin\big(\omega t + \angle G(j\omega)\big)$$

for input $u = \hat{u}\sin\omega t$. **Recipe**: replace s by $j\omega$.

Example $G = \dfrac{1}{s+1}$:

1. $\omega = 1$ rad/s: $G(j1) = \dfrac{1}{1+j} = 0.707\angle -45°$
2. $\omega = 10$ rad/s: $\dfrac{1}{1+10j} = 0.0995\angle -84°$

Faster input $\Rightarrow$ smaller output and more lag.

Refresher: decibels and logarithmic scales

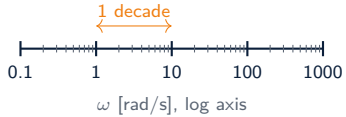
Refresher: the decibel

Gains span from 0.001 to 1000: a **log scale** keeps both visible. Gain in decibels:

$$|G|_{\text{dB}} = 20 \log_{10} |G|$$

- Gains in series multiply $\Rightarrow$ their dB values **add**.
- $-3\,\text{dB} = 0.707$: the usual edge of the "bandwidth".
- A **decade** = frequency $\times 10$.

$\lvert G\rvert$	dB
1000	$+60$
100	$+40$
10	$+20$
2	$+6$
1	0
0.707	-3
0.5	-6
0.1	-20
0.01	-40



Equal steps on the axis = equal *ratios*.

Example: $|G| = 0.05 \Rightarrow 20 \log_{10} 0.05 = -26\,\text{dB}$.

How to read a Bode plot

A **Bode plot** = frequency response for all ω: magnitude in dB (top) and phase in degrees (bottom), both over a log frequency axis.

Reading it:

1. pick ω (e.g. 0.1 rad/s)
2. read magnitude: 40 dB $\Rightarrow$ gain $\times 100$
3. read phase: $\approx -85^\circ$ (output lags)

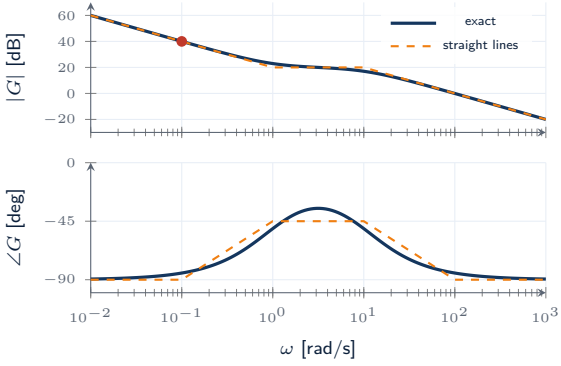
Straight-line rules:

- each pole: -20 dB/decade after its corner, phase drops by 90°
- each zero: $+20$ dB/decade, phase rises by 90°
- integrator $1/s$: -20 dB/decade, -90° everywhere



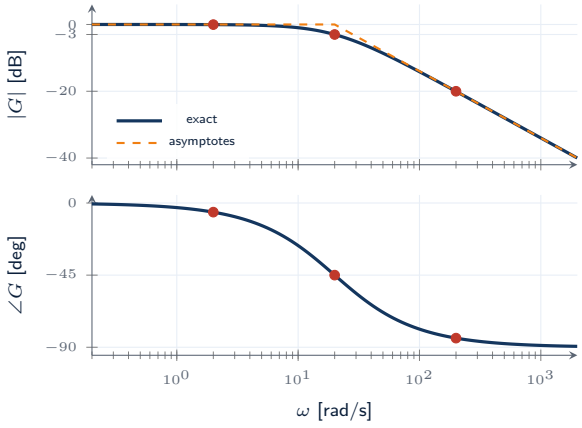
H. W. Bode (1905–1982), Bell Labs. Unknown author, Public domain, via Wikimedia Commons

$$G(s) = \frac{100(s+1)}{s(s+10)}$$



Integrator: -20 dB/dec; zero at 1 flattens; pole at 10 bends down again.

Worked example: Bode plot of a first-order lag



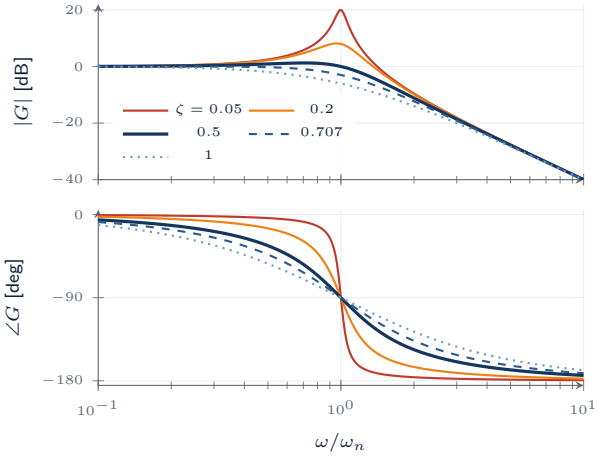
Current-controlled joint motor seen from the speed command, $\tau = 0.05\,\text{s}$:

$$G(s) = \frac{1}{\tau s + 1}, \quad \omega_b = \frac{1}{\tau} = 20 \text{ rad/s}$$

1. $\omega = 2$: $|G| = 1/\sqrt{1 + 0.01} = 0.995$ ($-0.04\,\text{dB}$), $-5.7°$.
2. $\omega = 20$ (corner): $1/\sqrt{2} = 0.707$ ($-3\,\text{dB}$), $-45°$.
3. $\omega = 200$: ≈ 0.1 ($-20\,\text{dB}$), $-84°$.

Two straight lines (flat, then $-20\,\text{dB/dec}$) are within 3 dB everywhere.

Resonance: pushing a swing at the right moment



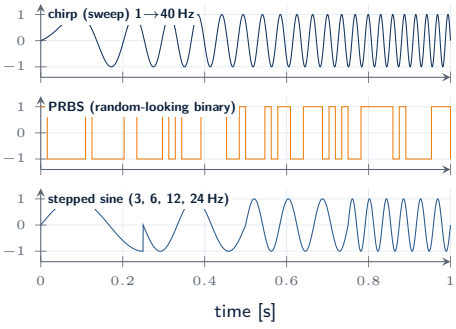
Small pushes at the swing's own rhythm give a **big** motion: that is **resonance**.

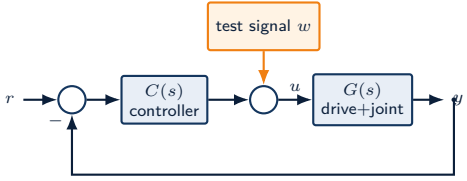
$$|G(j\omega_n)| = \frac{1}{2\zeta}$$

- Below ω_n: output follows ($0\,$dB, 0°).
- At ω_n: peak, phase -90°.
- Above ω_n: falls with $-40\,$dB/decade, phase $\to -180^\circ$.
- $\zeta = 0.05$: peak $1/0.1 = 10$ ($+20\,$dB) – typical of a **gearbox or belt** resonance in a robot joint.

Going further: peak $M_r = 1/(2\zeta\sqrt{1-\zeta^2})$ at $\omega_n\sqrt{1-2\zeta^2}$.

Measuring a frequency response on a robot axis

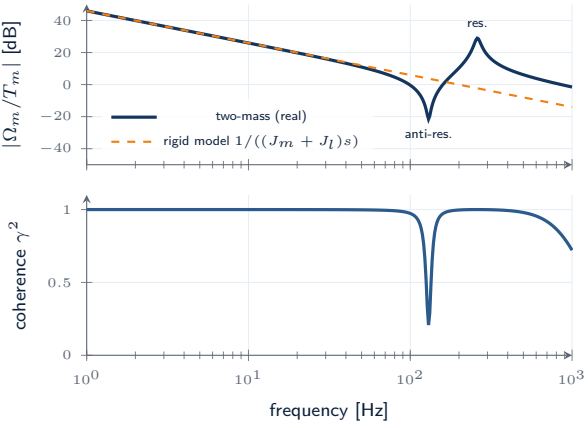




- The drive **adds** a test signal w while the joint is held in closed loop (safe against gravity).
- It records u and y and computes $|G|$ and $\angle G$ at each frequency.
- Amplitude: big enough to beat friction and noise, small enough to stay within limits.
- Average several runs to reduce noise.

Going further: FRF $\hat{G} = \hat{S}_{wy}/\hat{S}_{wu}$ from cross-spectra.

Measured response of a compliant joint: two masses and a spring



Motor $J_m = 2 \times 10^{-4}$ and load $J_l = 6 \times 10^{-4}$ kg m^2 joined by a springy gear/shaft $k = 400$ Nm/rad:

$$f_r = \frac{1}{2\pi}\sqrt{k\,\frac{J_m + J_l}{J_m J_l}} = 260 \text{ Hz}$$

- Up to ≈ 50 Hz the rigid model fits; near 130–260 Hz it is **completely wrong**.
- **Coherence** γ^2 $(0 \ldots 1)$: trust the measurement only where $\gamma^2 \gtrsim 0.8$.
- Rule: keep the control bandwidth **below the first resonance** (notch filters: W1D4).

Session 2.4 – Check your understanding

Check your understanding

1. Convert a gain of 0.1 and a gain of 1000 to dB.
2. At 10 Hz a 1 V input sine gives a 2 V output sine that lags by a quarter period. What are $|G|$ in dB and $\angle G$?
3. A robot joint with a strain-wave gear (right) shows a sharp peak at 40 Hz in its Bode plot. What does this mean for the controller?



Pieceofmetalwork, CC BY-SA 4.0, via Wikimedia Commons

Strain-wave (harmonic) gear set: light, zero backlash, but **springy**.

Session 2.4 – Summary and key terms

Key take-aways

- Sine in $\Rightarrow$ sine out: gain $|G(j\omega)|$ and phase $\angle G(j\omega)$.
- Bode plot: dB and degrees over log frequency.
- Each pole: $-20\,\text{dB/dec}$ and up to $-90°$; corner at $\omega = 1/\tau$ is $-3\,\text{dB}$.
- Low $\zeta \Rightarrow$ sharp resonance peak $1/(2\zeta)$.
- Measured Bode plots reveal joint resonances $\Rightarrow$ **limit the bandwidth**.

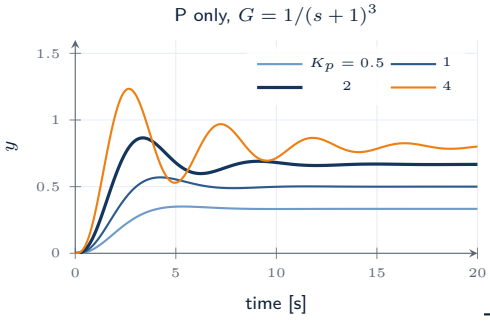
Key terms

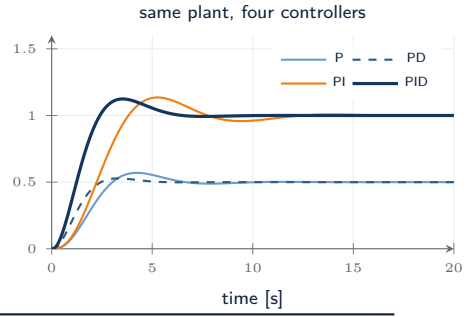
Frequency response: gain and phase vs. frequency.
Decibel: $20\log_{10}$ of a gain.
Corner frequency: where asymptotes meet.
Resonance: large response near ω_n.
Coherence: measurement quality, $0 \ldots 1$.

Session 3

2.5 PID Control

P, I and D: present, past and future of the error



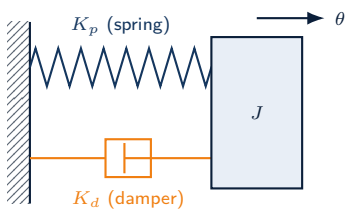


$$u(t) = K_p\, e + K_i \int_0^t e\, d\tau + K_d\, \dot{e}$$

P = now, **I** = accumulated past, **D** = trend (future).

More ...	speed	overshoot	steady-state error
P (K_p)	faster	larger	smaller, not zero
I (K_i)	faster	larger	**removed**
D (K_d)	similar	smaller	no effect (noise ↑)

PD control of a joint = a virtual spring and damper



spring anchored at the target θ_r

$$T = K_p(\theta_r - \theta) - K_d\,\dot{\theta}$$

D acts on the measured speed $\dot{\theta}$ (no kick on target jumps).

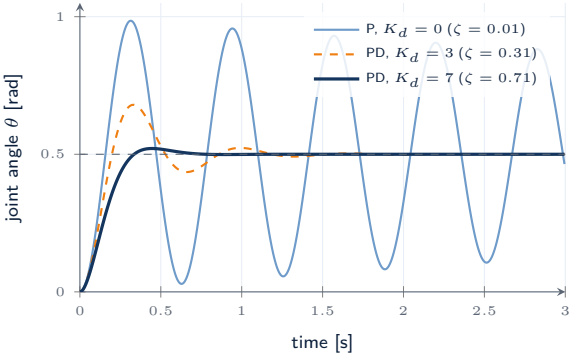
The controller makes the joint **feel** like a mass on a spring with a shock absorber – the car suspension from W1D2.

Worked example: choose the gains

Target: $\omega_n = 10\,\text{rad/s}$, $\zeta = 0.7$; joint $J = 0.5$, $B = 0.1$.

1. Stiffness: $K_p = J\omega_n^2 = 0.5 \cdot 100 = 50\,\text{Nm/rad}$.
2. Total damping $2\zeta\omega_n J = 2 \cdot 0.7 \cdot 10 \cdot 0.5 = 7\,\text{Nm s/rad}$.
3. $K_d = 7 - B = 6.9 \approx 7\,\text{Nm s/rad}$.
4. Expect $\approx 5\,\%$ overshoot, settling $4/(\zeta\omega_n) \approx 0.6\,\text{s}$.

Effect of D on the joint: from ringing to smooth

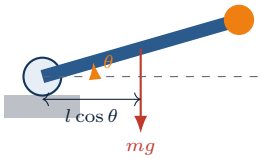


Step of 0.5 rad, $K_p = 50$ Nm/rad, simulated.

- Same $K_p \Rightarrow$ same stiffness and roughly same speed.
- K_d only adds **damping**: overshoot falls from 97 % to 4 %.
- Too much K_d: sluggish, and it amplifies encoder noise.

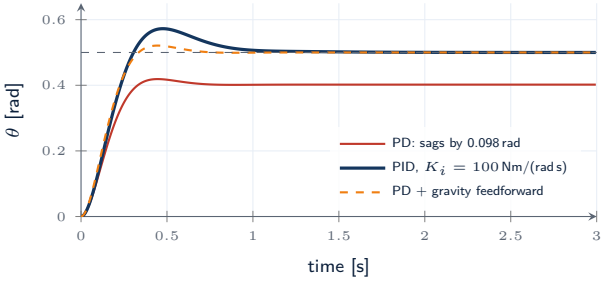
In a servo drive, the D-term on position is often realised by the **inner velocity loop** (W1D5).

Gravity: why the I-term is needed



$$T_g = m\,g\,l\cos\theta$$

$m = 1\,\text{kg}$ at $l = 0.5\,\text{m}$: $T_g \approx 4.9\,\text{Nm}$ near horizontal.



1. With PD at rest: $K_p\,e = T_g \Rightarrow e = 4.9/50 = 0.098\,\text{rad} = 5.6^\circ$.
2. The integrator keeps growing until the error is **exactly zero**; it "learns" the 4.9 Nm.
3. Gravity feedforward does the same instantly if m, l are known (session 2.9).

PID forms: same idea, different parameters

Form	Transfer function	Typical use
Parallel	$C = K_p + \dfrac{K_i}{s} + K_d s$	textbooks, drives
Ideal (ISA)	$C = K_p\left(1 + \dfrac{1}{T_i s} + T_d s\right)$	process controllers
Series	$C = K_p'\left(1 + \dfrac{1}{T_i' s}\right)\left(1 + T_d' s\right)$	older PLCs

Conversion parallel $\leftrightarrow$ ideal: $K_i = K_p/T_i$, $K_d = K_p T_d$.
T_i = integral time [s], T_d = derivative time [s].

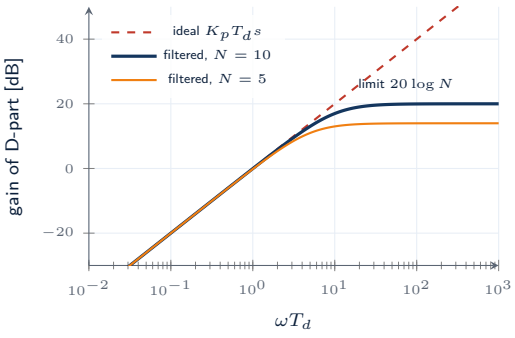
- Example: $K_p = 2$, $T_i = 4\,\mathrm{s}$, $T_d = 0.5\,\mathrm{s} \Rightarrow K_i = 0.5\,\mathrm{s}^{-1}$, $K_d = 1\,\mathrm{s}$.
- **Pitfall**: copying gains between devices without checking form and units (s vs. min, % vs. A).



Bitjungle, CC BY-SA 4.0, via Wikimedia Commons

Industrial PID controllers usually ask for K_p, T_i, T_d (ideal form).

Derivative action: always filter it



A pure derivative amplifies fast wiggles without limit: sensor noise becomes huge.

$$D(s) = \frac{K_p T_d\, s}{1 + T_d s/N}, \qquad N \approx 5 \ldots 20$$

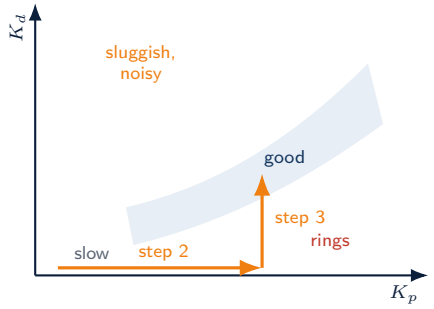
- The filter caps the high-frequency D-gain at $K_p N$.
- **Derivative on measurement** ($-\dot{y}$ instead of $\dot{e}$): no "kick" when the target jumps.
- Encoder example: 20-bit encoder, 125 μs sampling: one count gives a speed jump of 0.048 rad/s.

Tuning recipe 1: by hand on the robot joint

Step-by-step hand tuning

1. Start safely: low speed, small steps, $K_i = K_d = 0$, limits active.
2. Raise K_p until the response is fast but starts to ring.
3. Raise K_d until the ringing is gone ($\approx 5\,\%$ overshoot).
4. Add a little K_i to remove the gravity sag; stop when overshoot grows.
5. Re-check with the heaviest payload and in the worst pose.

Rule of thumb: halve any gain that makes the motor **sing** (audible high-frequency noise).



A stiffer joint (higher K_p) needs a proportionally larger K_d: keep ζ near 0.7.

Tuning recipe 2: Ziegler–Nichols (1942)

Recipe: use P only, raise K_p until the loop just keeps oscillating. Note this **ultimate gain** K_u and the period T_u.

	K_p	T_i	T_d
P	$0.5K_u$	–	–
PI	$0.45K_u$	$T_u/1.2$	–
PID	$0.6K_u$	$T_u/2$	$T_u/8$

Going further: model-based rules (SIMC, lambda tuning) choose a desired closed-loop time constant instead [Skogestad, 2003].

Worked example

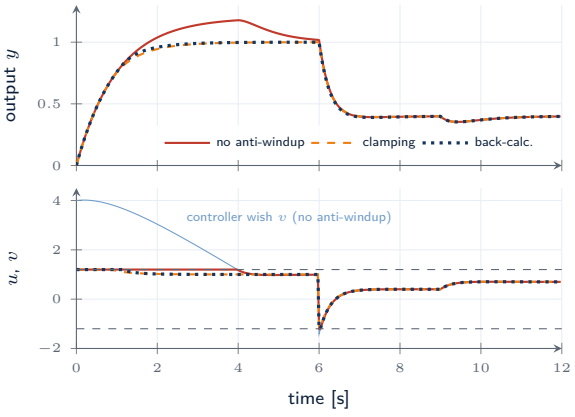
Plant $G = 1/(s+1)^3$.

1. Each pole lags up to $90°$; total $-180°$ when $3\arctan\omega = 180° \Rightarrow \omega_u = \sqrt{3} = 1.73\,\text{rad/s}$.
2. There $|G| = 1/(1+3)^{3/2} = 1/8 \Rightarrow K_u = 8$.
3. $T_u = 2\pi/1.73 = 3.63\,\text{s}$.
4. PID: $K_p = 4.8$, $T_i = 1.81\,\text{s}$, $T_d = 0.45\,\text{s}$.

Caution

Gives fast but **aggressive** loops. Pushing a real robot to oscillation can be unsafe: use ZN only as a starting point [Åström & Hägglund, 2006].

Saturation and integrator windup



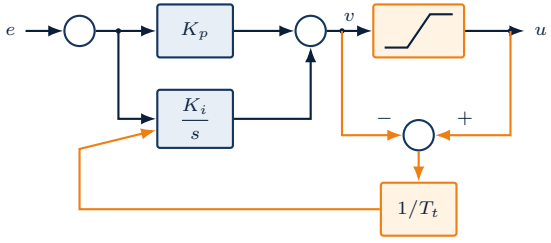
Plant $1/(s+1)$, PI $K_p = 4$, $T_i = 0.8\,\text{s}$, limit $|u| \le 1.2$.

Every real actuator has a **saturation**: the drive's current limit, the motor's torque limit.

- Joint blocked by an obstacle, or a long fast move: torque stuck at the limit.
- The integrator keeps adding up error although the torque **cannot** grow: it "winds up".
- When freed, it must first "unwind" ⇒ big overshoot (18 % here) and slow settling.

> A wound-up joint can **lunge** when the obstacle is removed – a safety issue.

Anti-windup: two standard cures



Clamping

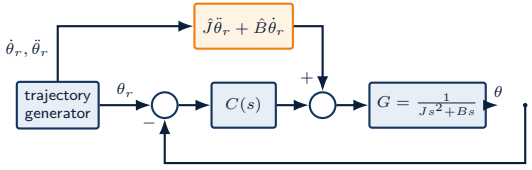
Stop integrating while the output is at its limit *and* the error pushes further. Simple; standard in servo drives.

Back-calculation

Feed the difference $u - v$ (actual minus wished) back into the integrator, scaled by $1/T_t$, so it tracks the limit. Typical $T_t \approx T_i/2$.

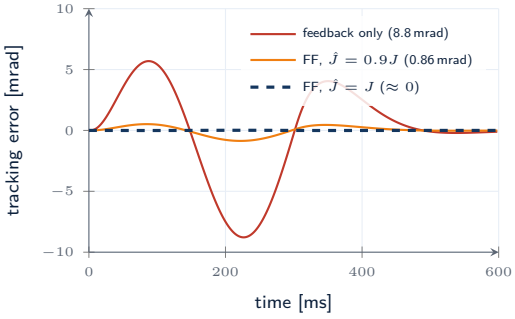
Every controller with an integrator and a saturating actuator needs one of these. **Going further:** in cascaded loops, pass the inner loop's limit to the outer integrator.

Feedforward: let feedback correct only what is unknown



A driver who *sees* the hill presses the pedal **before** the car slows: that is feedforward.

$$T_{\mathrm{FF}} = \hat{J}\,\ddot{\theta}_r + \hat{B}\,\dot{\theta}_r\ (+\hat{T}_g)$$

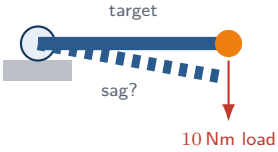


- Even with a 10 % inertia error: $10\times$ smaller error.
- FF sits outside the loop: it does **not** affect stability.
- Needs a smooth reference (S-curve profile, W1D5).

Session 2.5 – Check your understanding

Check your understanding

1. A PD-controlled joint ($K_p = 200\,\text{Nm/rad}$) holds a 10 Nm gravity load. How large is the sag? Which term removes it?
2. Why do we always filter the D-term?
3. A gripper closes on a part and is blocked for 2 s with the torque at its limit. What happens inside a PI controller without anti-windup?



Session 2.5 – Summary and key terms

Key take-aways

- P = virtual spring, D = virtual damper, I = learns constant loads (gravity).
- Pick $K_p = J\omega_n^2$, $K_d \approx 2\zeta\omega_n J$ as a starting point.
- Filter D; apply it to the measurement.
- Every saturating integrator needs **anti-windup**.
- Feedforward (inertia, friction, gravity) improves tracking without changing stability.

Key terms

PID: proportional–integral–derivative.
Integral time T_i, **derivative time** T_d.
Saturation: actuator at its limit.
Windup: integrator grows while saturated.
Feedforward: act on the known command before an error appears.

Session 4

2.6 Stability and Performance

What does “stable” mean?



stable
returns

unstable
runs away

marginal
stays where pushed

Shower analogy: the hot water arrives late (delay). If you over-react to what you feel now, the temperature swings ever more – **feedback can create instability**.

Closed-loop poles are the roots of the **characteristic equation** $1 + L(s) = 0$.

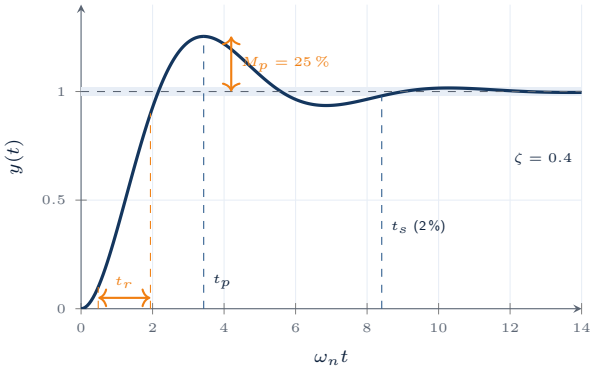
stable $\iff$ all poles: $\mathrm{Re}(p) < 0$

Example $L(s) = \dfrac{K}{s(s+2)(s+5)}$:

1. $1 + L = 0 \Rightarrow s^3 + 7s^2 + 10s + K = 0$
2. $K = 20$: poles -5.88 and $-0.56 \pm j1.76 \Rightarrow$ stable
3. $K = 100$: two poles in the right half-plane $\Rightarrow$ unstable

Too much gain destabilises – we will see why on the next slides.

Time-domain specifications



Measure	Meaning
Rise time t_r	$10\,\% \to 90\,\%$
Overshoot M_p	how far past the target
Settling time t_s	stays within $\pm 2\,\%$
Steady-state error	error left at the end

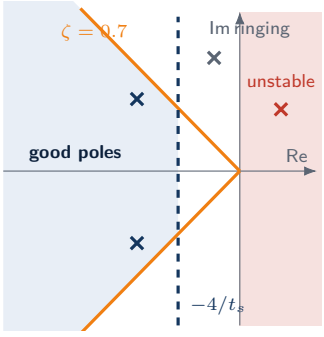
For a 2nd-order closed loop, ζ **alone fixes the overshoot**:

$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}, \qquad t_s \approx \frac{4}{\zeta\omega_n}$$

ζ	0.3	0.4	0.5	0.6	0.7
M_p [%]	37	25	16	9.5	4.6
PM [deg]	33	43	52	59	65

Robot pick-and-place: usually $M_p < 5\,\%$ (do not hit the table!).

From specifications to pole positions



Each spec draws a line in the s-plane; good poles sit in the **shaded wedge**.

- Overshoot $\Rightarrow \zeta \geq \zeta_{\min}$: inside the angle lines.
- Settling $\Rightarrow$ real part $\leq -4/t_s$: left of the dashed line.

Worked example

Joint must settle in $t_s = 0.4\,\text{s}$ with $< 5\,\%$ overshoot.

1. $\zeta \geq 0.7$; $\zeta\omega_n \geq 4/0.4 = 10\,\text{s}^{-1}$.
2. $\omega_n \geq 10/0.7 = 14.3\,\text{rad/s}$.
3. $K_p = J\omega_n^2 = 0.5 \cdot 204 = 102\,\text{Nm/rad}$.

Steady-state error and system type

Final value theorem (only for a **stable** loop):

$$e_\infty = \lim_{t\to\infty} e(t) = \lim_{s\to 0} s\,E(s), \qquad E = \frac{R}{1+L}$$

System type = number of integrators $(1/s)$ in $L(s)$.

Type	step	ramp	example
0	$\frac{1}{1+L(0)}$	∞	cruise control, P only
1	0	$\frac{1}{K_v}$	PI, or position loop
2	0	0	position loop + I

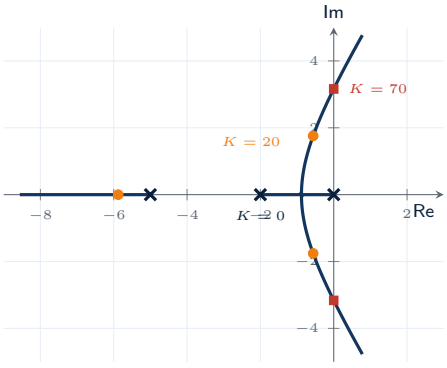
$K_v = \lim_{s\to 0} sL(s)$ [s^{-1}], the velocity constant.

Worked example: gantry axis following error

1. Position P gain $K_{pp} = 126\,\mathrm{s}^{-1}$ around a fast speed loop: $L \approx K_{pp}/s \Rightarrow$ type 1, $K_v = 126\,\mathrm{s}^{-1}$.
2. Axis moves at 0.2 m/s, 10 mm ball-screw pitch $\Rightarrow$ motor speed $\dot\theta = 126$ rad/s (a ramp in θ).
3. $e_\infty = \dot\theta / K_v = 126/126 = 1$ rad.
4. $1\,\mathrm{rad} \times 10\,\mathrm{mm}/2\pi = 1.6\,\mathrm{mm}$ lag behind the command.

Cure: velocity feedforward (session 2.5) removes this lag. Gravity on a P-only joint is a type-0 **step disturbance**.

Root locus: how closed-loop poles move with gain

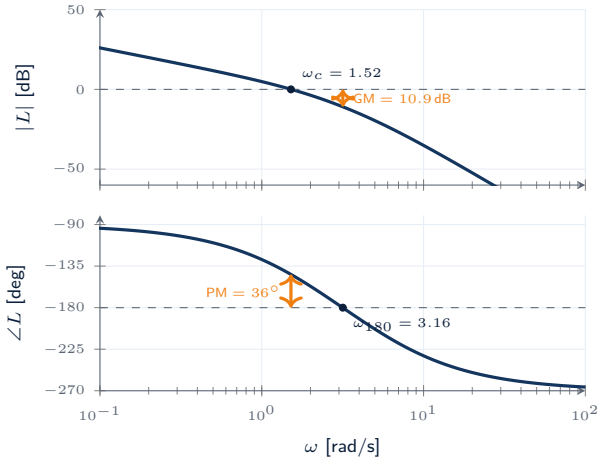


Same loop $L = \dfrac{K}{s(s+2)(s+5)}$; the curves show where the 3 closed-loop poles go as K grows from 0.

- $K = 0$: poles start at the open-loop poles $0, -2, -5$ ($\times$).
- Small K: two poles meet and split into a complex pair $\Rightarrow$ oscillation starts.
- $K = 20$: $-0.56 \pm j1.76$ ($\zeta = 0.30$, $\approx 37\,\%$ overshoot).
- $K = 70$: poles reach the $j\omega$-axis at $\pm j3.16$ $\Rightarrow$ **stability limit**.
- $K > 70$: unstable.

Going further: the Routh–Hurwitz table gives the same limit $0 < K < 70$ from the coefficients alone; the Nyquist plot of $L(j\omega)$ must not encircle -1.

Gain and phase margins: how far from the edge?

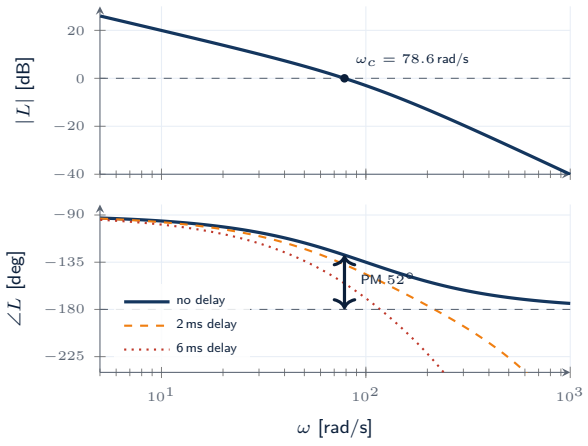


The loop oscillates if a signal comes back around the loop with gain 1 and $-180°$. Margins = **safety distance** from that point.

1. Find the **crossover** ω_c where $|L| = 0\,\text{dB}$ ($1.52\,\text{rad/s}$).
2. Read the phase there: $-144° \Rightarrow$ **phase margin** PM $= 180° - 144° = 36°$.
3. Find where the phase is $-180°$ ($3.16\,\text{rad/s}$); $|L| = -10.9\,\text{dB}$ there $\Rightarrow$ **gain margin** GM $= 10.9\,\text{dB}$ ($\times 3.5$).

- Same loop as the root locus with $K = 20$: GM $\times 3.5 \Rightarrow K = 70$ is the limit. **Consistent!**
- Targets: PM 45–$60°$, GM $\geq 6\,\text{dB}$.

Worked example: margins of a joint position loop

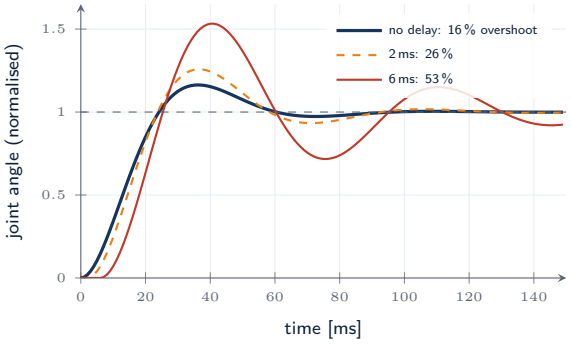


Joint position loop (speed loop inside, lag $\tau = 10\,\text{ms}$):

$$L(s) = \frac{100}{s\,(0.01s + 1)}$$

1. Crossover: $|L| = 1 \Rightarrow \omega_c \approx 78.6\,\text{rad/s}$ (12.5 Hz).
2. Phase: $-90° - \arctan(0.786) = -128.2° \Rightarrow \text{PM} = 51.8°$.
3. Phase never reaches $-180° \Rightarrow$ GM is infinite.
4. A delay T_d removes $\omega_c T_d$ rad: 2 ms $\Rightarrow -9°$ (PM 43°); 6 ms $\Rightarrow -27°$ (PM 25°).

Time delay: the shower problem in a robot



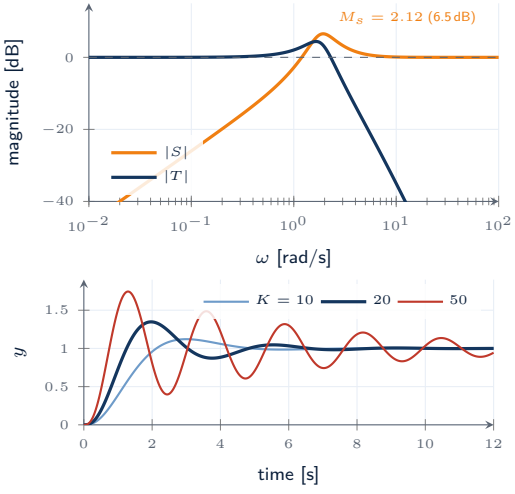
Same loop as previous slide, closed with unity feedback; simulated.

Delays come from sampling, computation, filters and **networks** (EtherCAT, CAN: W2D5).

$$\Delta\varphi = -\omega_c\, T_d \text{ [rad]}$$

- A delay changes **phase only**, not gain: it cannot be seen in $|L|$.
- Limit here: $0.904\,\text{rad}/78.6\,\text{rad/s} = 11.5\,\text{ms}$ of delay makes the joint oscillate.
- The faster the loop (higher ω_c), the less delay it tolerates.

Sensitivity, robustness and loop shaping



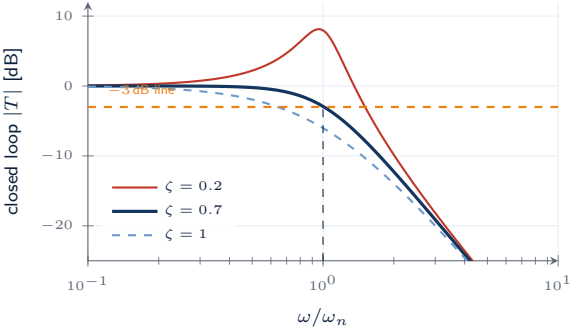
$|S(j\omega)|$ = how much of a disturbance at frequency ω survives the loop.

$$M_s = \max_{\omega} |S(j\omega)| \quad \text{(aim: } 1.2 \ldots 2\text{)}$$

- Low frequency: $|S| \ll 1 \Rightarrow$ disturbances removed.
- Near crossover: $|S| > 1 \Rightarrow$ feedback **amplifies** disturbances.
- $K = 20$: $M_s = 2.1$, PM $36^\circ \Rightarrow$ ringing; $K = 10$ is calmer.
- **Loop shaping**: make $|L|$ big at low frequency (tracking), small at high frequency (noise, resonances).

Going further: waterbed effect: pushing $|S|$ down at some frequencies pushes it up elsewhere.

Bandwidth: how fast can the joint follow?



Bandwidth ω_{bw}: highest frequency the closed loop still follows, $|T| \geq -3\,\text{dB}$.

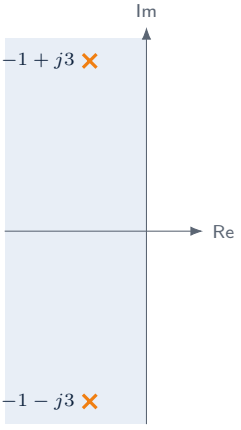
$$t_r \approx \frac{2.2}{\omega_{bw}} = \frac{0.35}{f_{bw}}$$

- Rule of thumb: $\omega_{bw} \approx (1 \ldots 1.6)\,\omega_c$.
- Our joint loop: $\omega_c = 78.6\,\text{rad/s}$, $\omega_{bw} \approx 127\,\text{rad/s}$ ($20\,\text{Hz}$) $\Rightarrow t_r \approx 17\,\text{ms}$.
- Higher bandwidth = faster, but limited by **resonances** ($\omega_{bw} < \omega_r/3$), delay and noise.

Session 2.6 – Check your understanding

Check your understanding

1. Closed-loop poles at $-1 \pm j3$. Stable? Small or large overshoot?
2. A loop has a phase margin of only 10°. What would you see in its step response?
3. A type-1 position loop has $K_v = 100\,\mathrm{s}^{-1}$ and moves at 50 rad/s. What is the following error?
4. Crossover is 100 rad/s, PM $= 40^\circ$. How much delay brings it to the edge?



Session 2.6 – Summary, design targets and key terms

Key take-aways

- Stable $\iff$ closed-loop poles in the left half-plane.
- ζ sets overshoot; PM $\approx 100\zeta$.
- System type sets the steady-state error.
- Delay eats phase: $\omega_c T_d$.
- Bandwidth sets speed: $t_r \approx 0.35/f_{bw}$.

Typical design targets

Quantity	Target
Phase margin	45–60°
Gain margin	≥ 6 dB
M_s	1.2–2.0
Overshoot (position)	0–5 %
ω_{bw} vs. resonance	$< \omega_r/3$

Key terms

Characteristic eq. $1 + L = 0$.
Crossover ω_c: $|L| = 1$.
Phase margin: extra lag tolerated.
Gain margin: extra gain tolerated.
System type: integrators in L.
Bandwidth: -3 dB of T.

Session 5

Wrap-up

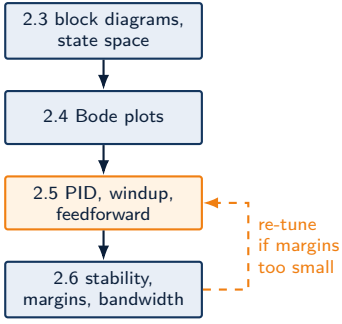
Day summary

The control design chain

model $G(s)$ or (A, B, C, D) $\rightarrow$ block diagram with $C(s)$ $\rightarrow$ Bode plot of $L = CG$ $\rightarrow$ pick PID gains $\rightarrow$ check poles, PM, GM, bandwidth $\rightarrow$ add anti-windup and feedforward.

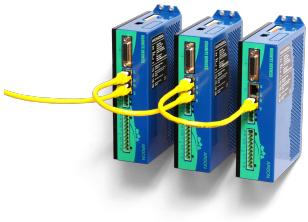
Numbers to remember

$-20\,\text{dB/dec}$ per pole ▪ $-3\,\text{dB}$ at the corner ▪ $\zeta = 0.7$: 5 % overshoot ▪ PM 45–60° ▪ GM $\geq 6\,\text{dB}$ ▪ $t_r \approx 0.35/f_{bw}$



Preview of W1D4: control (cont.) and servo systems I

- **2.7** Filters and compensation: low-pass, notch (for today's resonance), lead/lag.
- **2.8** Digital control: sampling, zero-order hold, discrete PID, delay.
- **2.9** Control of robot manipulators: independent joints, gravity compensation, computed torque.
- **3.1** Servo architecture: torque / velocity / position modes.
- **3.2** Power electronics for drives: DC bus, inverter, PWM.



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Servo drives: today's PID loops live inside these boxes.

Figure credits

All other drawings and plots: own work (TikZ / circuitikz / pgfplots; numerical data in `w1d3/data/`).

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d4_kuka_robot.jpg	File:KUKA industrial robotic arm performing “The Beach” by Roy Nachum - 02.jpg	Oleg Yunakov	CC BY-SA 4.0
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d2_pid_controllers.jpg	File:Industrial PID controllers - front display.jpg	Bitjungle	CC BY-SA 4.0
d2_servo_drives.png	File:Servo motor drive.png	Tero Kontkanen	CC BY-SA 3.0

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